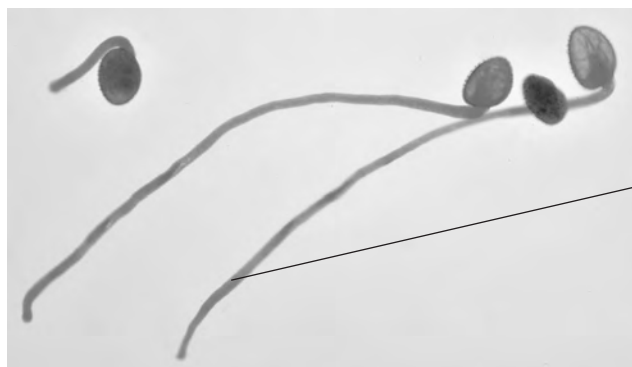


- 4 The photograph shows some pollen tubes growing from pollen taken from a lily flower, as seen using a light microscope.



Magnification $\times 100$

(Source: © agefotostock/Alamy Stock Photo)

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- (a) (i) The pollen tube growth rate is $0.2 \mu\text{m s}^{-1}$.

How long would it take the pollen tube to grow 15 mm?

(1)

- ☐ **A** 75 seconds
- ☐ **B** 125 minutes
- ☐ **C** 1 250 minutes
- ☐ **D** 7 500 seconds

- (ii) State how to calculate the actual length of one of the pollen tubes in the photograph.

(1)

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A black and white photograph of a lily flower. The flower is in full bloom, showing six large, light-colored petals with prominent veins. The center of the flower reveals six stamens with dark, rounded anthers. Above the open flower, two unopened buds are visible on the same stem, pointing upwards. The background is dark and out of focus, making the flower stand out.

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(Total for Question 4 = 11 marks)

